Tutorial 3: Electronic structure of atoms

Exercise 1. ***(Know)***

Write the full and condensed electron configurations for the following atoms, using the appropriate noble-gas core abbreviations: (a) C, (b) O, (c) P, (d) Cl, (e) Na, (f) Ca.

Exercise 2. ***(App)***

The following electron configurations represent excited states. Identify the element and write its ground-state condensed electron configuration.

- **(a)** $1s^2 2s^2 2p^4 3s^1$
- **(b)** $1s^2 2s^2 2p^6 3s^1 3p^6 4s^2 4p^1$
- **(c)** $[Ne] 3s^1 3p^5 4p^1$
- **(d)** $[Ar] 4s^2 3d^9 4p^3$

Exercise 3. ***(Und)***

The contour representation of one of the orbitals for the n = 3 shell of a hydrogen atom is shown here.

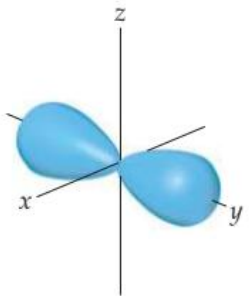
(a) What is the quantum number l for this orbital?
(b) How do we label this orbital?
(c) In which of the following ways would you modify this sketch if the value of the magnetic quantum number, m_l, were to change?

(i) It would be drawn larger,

(ii) the number of lobes would change,

(iii) the lobes of the orbital would point in a different direction,

(iv) there would be no change in the sketch



Exercise 4. ***(Know)***

Sketch the shape and orientation of the following types of orbitals:

- **(a)** s
- **(b)** p_x
- **(c)** p_z
- **(d)** d_{xy}
- **(e)** d_{xy}
- **(f)** $d_{x2\text{-}y2}$

Exercise 5. ***(Und)***

There are sets of valid quantum numbers for the hydrogen atom. Indicate which of the quantum number set is valid or not valid. If it is invalid, provide the valid one.

(a) $n = 3, l = 3, m_l = 2, m_s = +½$
(b) $n = 3, l = 3, m_l = -3, m_s = +1/2$
(c) $n = 3, l = 1, m_l = 2, m_s = +1/2$
(d) $n = 5, l = 0, m_l = 0, m_s = 0$
(e) $n = 2, l = 1, m_l = 1, m_s = -1$

Exercise 6. ***(App)***

Known that the speed of light is $2.998 * 10^8$ m/s.

(a) A laser used in orthopedic spine surgery produces radiation with a wavelength of 2.10 μm. Calculate the frequency and energy of this radiation.

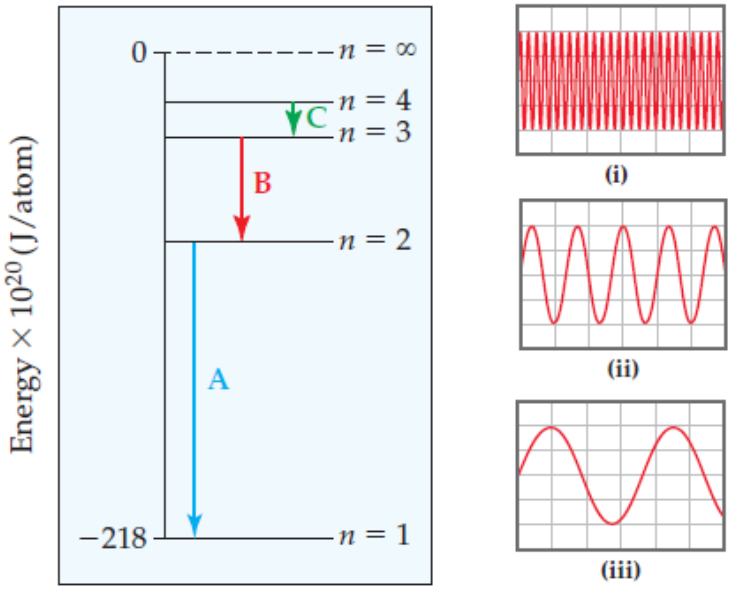
(b) What are the wavelength and energy of another laser that has a frequency of 5.50×10^{14} s^{-1}?

(c) Would the radiation in part (a) or part (b) be visible to the human eye?

Exercise 7. ***(App)***

Consider the three electronic transitions in a hydrogen atom shown here, labeled A, B, and C.

(a) Three electromagnetic waves, all drawn on the same scale, are also shown. Each corresponds to one of the transitions. Which electromagnetic wave (i), (ii), or (iii), is associated with electronic transition C?



(b) Calculate the energy of the photon emitted for each transition.

(c) Calculate the wavelength of the photon emitted for each transition. Do any of these transitions lead to the emission of visible light? If so which one(s)?

Exercise 8. ***(Adv App)***

The photons of the infrared radiation are responsible for warming other objects. Known that 2.2×10^{23} photons are absorbed by the water to warm a cup of water (150 g) from 25.0 °C to 55.0 °C, the specific heat capacity of water is 4.184 J g^{-1} $°C^{-1}$. What is the wavelength of infrared photons emitted from fire?